

Technology Stack

Date	24 June 2026
Team ID	SWTD-2026-3249
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the OptiCrop solution. The chosen technology stack ensures the application is scalable, maintainable, and aligned with the project's requirements for real-time agricultural data processing and farmer accessibility.

The table below details the specific technologies chosen for each component of the architecture along with a brief justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js (web dashboard) + Flutter (farmer mobile app)	React provides a fast, component-reusable admin/agronomist dashboard, while Flutter delivers a single cross-platform codebase for Android/iOS with offline-first support — essential for farmers in low-connectivity rural areas.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python (Django REST Framework)	Python's mature data-science and ML ecosystem (NumPy, Pandas, scikit-learn) integrates naturally with Django REST Framework to serve the crop-optimization and recommendation models through clean, well-documented APIs.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	PostgreSQL with PostGIS + MongoDB	PostgreSQL/PostGIS handles structured farmer, farm, and crop records along with geospatial field-boundary queries, while MongoDB stores high-volume, semi-structured sensor and weather time-series data efficiently.

4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	AWS (EC2, S3, RDS) with Docker containers	AWS offers elastic, pay-as-you-grow compute for ML inference workloads and reliable managed storage/database services, while Docker ensures consistent environments across development, staging, and production.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	GitHub + GitHub Actions	GitHub enables seamless team collaboration and code review, while GitHub Actions automates testing, linting, and deployment pipelines, reducing manual release effort and catching issues early.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	OpenWeatherMap API, Twilio SMS API, Google Maps API	OpenWeatherMap feeds real-time weather data into irrigation/fertilizer recommendations, Twilio delivers SMS alerts to farmers without smartphones, and Google Maps enables accurate field location mapping and boundary tracing.